

2025 Ch H1 Q11

Section: Nature's Chemistry

Topic: Soaps, Detergents and Emulsions

Question summary

In an emulsion formed by shaking vegetable oil and water with an emulsifier, which part of the emulsifier dissolves in the oil droplets?

Worked solution

An emulsifier molecule has two parts: a hydrophobic (non polar) tail and a hydrophilic (polar) head.

The hydrophobic tail is attracted to and dissolves in the oil droplets.

The hydrophilic head is attracted to water, remaining in the aqueous layer.

This dual nature allows emulsifiers to stabilise oil and water mixtures.

Final answer

D. hydrophobic and dissolve in vegetable oil droplets

Revision tips

- Hydrophobic means 'water fearing' and refers to non polar groups that dissolve in oils.
- Hydrophilic means 'water loving' and refers to polar groups that interact with water.
- Emulsifiers stabilise emulsions by preventing oil droplets from merging back together.